Model Development and Final Evaluation

Technical report Part 2 | Bui Minh Hoang | 3 November 2026

1 Evaluation scope and frozen protocol

This part reports model development, a frozen final comparison and the resulting deployment decision for English ingredient-based recipe retrieval. It complements Part 1, which covers problem formulation, EDA, preprocessing and engineered features. The candidate corpus contains 50,000 RecipeNLG recipes. The system ranks existing recipes; it does not predict cooking time, nutrition, allergy safety or whether a particular person will enjoy a dish. The final result should therefore be read as evidence about a defined retrieval task, not a general guarantee of culinary usefulness.

Every evaluated system uses the same grouped outer split and seed 42. Near-duplicate groups remain within one split. The outer test partition is approximately 20% of recipes; validation is held out within the remaining outer-training partition. Text vectorizers and fine-tuning positives use training recipes. All 50,000 recipes remain retrieval candidates, including held-out origin recipes. This candidate-corpus convention makes the task a retrieval benchmark rather than ordinary classification. BM25 also computes index statistics over the candidate collection; this transductive convention is disclosed rather than described as a train-only vocabulary.

Synthetic queries are ingredient subsets of known recipes. Their relevance judgments identify origin recipes and any matched group positives. Recall@5 and Recall@10 measure recovery of these positives; MRR@10 measures the reciprocal rank of the first positive within ten results. A non-origin recipe can still be useful to a cook, so these scores are origin-recipe proxies. They do not replace the separate 0/1/2 human study. Empty retrieval results contribute zero rather than disappearing from the mean.

The team decision in this chat froze E01, E02, E03, LSA, E04, E05 and the CPU one-epoch FT checkpoint. BM25 (E03) was selected using validation before opening E10 scores. The protocol fixes top-k=10 and no missing-first reranking. It records hashes of corpus, splits, queries, qrels, configurations, model files, evaluation code and pinned pretrained encoder snapshot. The FT query encoder is packaged with its corpus embeddings and checked against file hashes.

E10 ran once on 1,000 main test queries. Parent MLflow run: 16e4dfb977434b2a87a06418c8f2e5a1. The exclusive execution ledger was written before test queries were loaded. A repeat invocation is rejected, including after failure; per-system rankings are preserved. Report generation reads those rankings and scores only. No hyperparameter, reranking or deployment choice was changed after test inspection. Hash and timestamp evidence is in reports/final/frozen_config.json and e10_execution.json.

2 Models and training procedure

TF-IDF represents each recipe as a sparse vector of canonical ingredient names. Multi-word ingredients remain atomic tokens, so chicken stock and chicken breast are distinct. The vocabulary and inverse document frequencies are fitted on training recipes, while every candidate recipe is transformed with that fitted vectorizer. Query and document vectors use cosine similarity. E01 is the default model; E02 uses parameters selected by grouped cross-validation. Serialization stores the fitted pipeline and corpus alignment, allowing the serving code to load a bundle without refitting it.

BM25 is a lexical ranking baseline implemented with rank-bm25. It combines ingredient overlap with inverse document frequency, term-frequency saturation and recipe-length normalization. The indexed candidate collection supplies its document statistics. This differs from TF-IDF's train-only fit, so the report names its timing as index construction rather than supervised training. BM25 is attractive when rare ingredient names sharply identify recipes; common or short ingredient queries remain ambiguous. A query with no ingredient overlap yields no lexical recommendation.

LSA applies TruncatedSVD to the train-fitted TF-IDF representation. The evaluated model uses 256 components, not the 100 dimensions mentioned in an older draft. The low-dimensional representation can connect co-occurring ingredients but may lose the exact rare terms needed to recover an origin recipe. It is evaluated as a separate model, not assumed superior because its representation is dense.

E04 and E05 use sentence-transformers/all-MiniLM-L6-v2, a six-layer transformer with 384-dimensional sentence embeddings. Text is truncated at 128 tokens in this project, pooled and L2-normalized. Queries contain normalized ingredients. E04 embeds ingredients only; E05 embeds title plus ingredients. Cosine search is implemented with sentence-transformers utilities and deterministic recipe-ID tie breaking. The encoder is pinned to the local snapshot recorded in the frozen manifest. These pretrained baselines have no training time on our recipe dataset; corpus encoding is a separate cost.

The FT system is the main-data E08 checkpoint, not pilot E07. It uses the ingredients-only format and MultipleNegativesRankingLoss on 30,000 train-only positive pairs. The historical measured run used one epoch, batch 32, learning rate 2e-5, weight decay 0.01 and 50 warmup steps on CPU. Observed training duration was 871.336 seconds. The loss treats other positives in a batch as negatives. The existing loader does not enforce the available no-duplicate group sampler, so same-group false negatives are a known limitation. This report evaluates the locked checkpoint without retraining it or claiming a controlled ablation.

3 Tuning and bias variance analysis

TF-IDF tuning used GridSearchCV with GroupKFold(5) on outer-training recipes. Each fold refitted the vectorizer using only its training groups. A held-out query was scored in the fold holding its origin recipe, and held-out recipes remained candidates. The grid compared whole-ingredient tokens with ingredient-plus-word tokens, min_df in {1,2,5}, sublinear term frequency and normalization choices. There were 24 candidate configurations. Tests verify that held-out-only terms do not enter a fold's fitted vocabulary. Using GroupKFold addresses group leakage; it does not eliminate all corpus bias.

The selected TF-IDF settings equal E01 defaults: whole-ingredient tokens, min_df=1, sublinear_tf=false and L2 normalization. CV MRR@10 was 0.6896 with standard deviation 0.0063. E02 therefore has the same Train, Validation and Test rankings as E01. This is a valid negative tuning result. The report does not invent a gain or describe E02 as a sublinear-TF model. Splitting multi-word names can increase broad matches, while a larger minimum document frequency removes rare identifiers; the CV results, rather than those intuitions alone, determine the selected setting.

LSA's grouped five-fold score is 0.3560 ± 0.0095 at the evaluated 256-component setting. This is a measured fixed-setting CV result, not a searched optimum. BM25 has no supervised fit in this setup, and the pretrained encoders are not fitted on recipe labels, so their CV entries are N/A with that reason. FT has one fixed checkpoint and no cross-validated retraining. N/A is not silently replaced with zero and must not be used in an average over model quality.

Train-proxy versus validation MRR@10 is 0.6998 versus 0.6567 for TF-IDF, 0.7687 versus 0.7583 for BM25, and 0.6450 versus 0.6209 for FT. These retrieval gaps are descriptive; train-proxy retrieval scores are not training losses. The single FT run and seed do not estimate training variance. A small observed gap cannot establish absence of overfitting, nor does a better test score justify retrospectively choosing a checkpoint.

The validation error analysis provides more specific evidence: 28 deterministic cases compare top-five results from all seven systems. It separates short queries, common-ingredient share, rare terms, singular/plural variants and large origin recipes. FT improved over pretrained E04 on 196 validation queries, regressed on 30 and tied on 274. These are cases from a fixed validation comparison, not new test-derived tuning rules. No test-case inspection was used to modify the frozen systems.

Controlled regularization evidence remains missing. The older ablation image was generated from literal loss arrays; it cannot support a causal claim about weight decay. A genuine comparison would require otherwise matched runs and measured loss/validation histories. That task belongs to Hiếu and is recorded in the rubric checklist; it is not filled in by this report.

4 Master Comparison Table and final results

All rows below use the frozen main corpus. Train scores are historical retrieval proxies over 5,000 train queries; Test scores come from the single E10 execution over 1,000 test queries. CV scores use grouped folds of the outer-training set. The asterisk marks the deployment decision made on validation before E10. It is not assigned by sorting this test table.

System	Train MRR	Test MRR	CV MRR	Train/index s	Tuned	Notes
E01	0.6998	0.6649	0.6896 ± 0.0063	28.024	No	Defaults
E02	0.6998	0.6649	0.6896 ± 0.0063	33.988	CV5	CV best=default
E03 *	0.7687	0.7766	N/A	25.658	No	Selected pretest; index
LSA	0.3402	0.3424	0.3560 ± 0.0095	51.638	No	256 components
E04	0.4575	0.4752	N/A	0.000	No	Ingredients
E05	0.1808	0.1989	N/A	0.000	No	Title + ingredients
FT	0.6450	0.6368	N/A	871.336	MNRL	E08; one CPU epoch

Notes: E01 and E02 have identical selected parameters. LSA's CV is a fixed-setting assessment, not tuning. BM25's time is index build; TF-IDF/LSA time includes fit and corpus transform. FT's duration is the historical CPU fine-tuning run. E04/E05 show zero dataset training seconds because their weights are pretrained; their nonzero corpus-encoding durations are stored separately. Different machines and timing definitions prevent a fair speed ranking from this training-time column.

BM25 test Recall@5 is 0.8600, Recall@10 is 0.9090, and MRR@10 is 0.7766. FT test MRR@10 is 0.6368, compared with 0.4752 for pretrained E04 and 0.1989 for E05. The result supports the frozen lexical deployment choice on this proxy; it does not establish that lexical systems satisfy every natural-language cooking intent.

The significance analysis uses ranx.compare and a two-sided paired Student t-test over per-query scores, $p < 0.01$. For test MRR@10, BM25 is significantly better than: E01 E02 E04 E05 FT LSA. The full per-metric comparisons and per-query scores are saved alongside the table. Pairwise tests are not corrected for multiplicity; the threshold must not be interpreted as a familywise error guarantee. E01/E02's identical ranks may produce undefined pairwise p-values rather than a win.

The final scores, raw test rankings and full MLflow run IDs are in reports/final. Table generation checks each copied Train/Validation/Test metric against its source run. It does not score a retriever again. The ledger records completion of all seven systems, and the selected model remains E03.

5 Deployment choice and practical limits

The selection rule was highest validation MRR@10 among the fixed systems, with performance and operational limitations disclosed. BM25's validation score of 0.7583 exceeded TF-IDF's 0.6567 and FT's 0.6209. The user confirmed E03 before E10. The production configuration now identifies the same BM25 main bundle; this is the configuration decision for Hưng to verify in the final container. This branch does not claim a new Docker build or a clean-clone deployment test.

BM25's serialized joblib bundle is small and does not require loading transformer weights for inference. The serving benchmark is a separate experiment: 20 warmup queries and 100 sequential validation queries per bundle, with startup, latency, RAM and artifact size recorded. These values describe one machine and workload. They cannot prove an end-user SLA, concurrent throughput or cold-start performance on a different host. E10 mean query latency also should not be substituted for the dedicated benchmark's p50/p95 measurements.

The underlying labels reward origin recovery. A different recipe containing the requested ingredients may be a better real meal than the origin. Conversely, ingredient overlap can reward recipes that violate an explicit intent such as vegetarian or no oven. The API has no reliable dietary constraint model, personalization, nutrition or availability knowledge. Human judgments must therefore consider intent, extra ingredients and usability rather than reproduce the proxy.

Preprocessing normalizes ingredient names and singular/plural variants, but parsing errors, ambiguous synonyms and specific ingredient substitutions remain. Treating stock as meat, ignoring a preparation requirement or merging distinct ingredients can mislead retrieval. The EDA audit and error-analysis cases document these issues; a high average MRR does not remove them. Deterministic recipe IDs, manifests and tie breaking make results reproducible, not automatically correct.

The model freeze includes artifact and encoder hashes and code provenance. The command refuses changed hashes or an existing E10 ledger. This supports auditability within this repository. It is not an enforcement mechanism against someone deleting audit files or using a separate tracking store. If a model must change after final test inspection, disclose the deviation and evaluate on newly reserved data rather than treating another pass as the same untouched test.

Potential future work includes an independently labelled evaluation set, better intent and substitution handling, safer no-overlap responses and multiple training seeds. These are proposals, not features implemented or gains demonstrated here. They should be chosen using future development data, not by tuning to the final query pool or revisiting E10 failures.

6 Human evaluation and ethical considerations

The natural-language study uses the existing 20 final queries, separate from the 20 development queries. The frozen systems contribute their top-five recipes per query; duplicates are merged and item order is shuffled with seed 42. Assigned CSVs contain query, recipe, ingredients and directions, but no model identity or rank. Queries are allocated round-robin to Hoàng, Hưng and Hiếu, with at least 20% of each query's items independently assigned to a second person.

The original annotation guide defines relevance 0/1/2 before the final pool is generated. Judges must work independently, avoid suggested AI labels and avoid system keys. Each person records actual timing, source and exposure in annotation_log.csv. A declaration is not signed by another person or filled by this report. Independent originals are hash-locked before disagreements are shown. Discussion is recorded in a separate adjudication table; original labels remain unchanged.

Current final-study status: awaiting_independent_labels. The pool and scoring implementation are ready, but independent final human labels have not yet been supplied. This report therefore publishes no human kappa or pooled nDCG score. It does not reinterpret blank labels as zero relevance, reuse dev judgments as final judgments, or present AI-generated judgments as agreement between the three team members.

After all originals are complete, the scoring command computes pairwise unweighted Cohen's kappa and raw agreement on the independent overlap. It then uses adjudicated labels to compute ranx nDCG@10 with linear relevance gains. Items outside the pool remain unjudged. The pooled calculation assigns them zero gain, a scoring convention that does not prove irrelevance. Every system is reported with mean and per-query top-ten judging coverage. A query with no positive pooled label and an undefined kappa denominator are explicitly limited cases, not flattering estimates.

Allergy and dietary safety require particular care. The source corpus does not provide verified allergen or cross-contamination information, and normalization can erase distinctions. Users should check the original ingredient list and personal restrictions; the system must not claim that a recommendation is safe for an allergy or a medical diet. The report's evaluation does not certify those properties. Deploying such a promise would require additional data, validation and controls.

Recipe sources and language coverage can bias the candidate collection. English-only recipes, popular ingredients and duplicated source styles may dominate the learned representation and judgment pool. The dataset licence has not been assumed or reassigned: cite RecipeNLG and source links, check redistribution rights and keep raw data/model artifacts out of source control. Prediction logs should be minimized and retained deliberately because ingredient histories can reveal personal habits. These limits travel with the reported scores.

7 Provenance and rubric review

The table below gives the full MLflow run IDs for all copied retrieval scores. Resolve them in the local sqlite:///mlflow.db tracking store, experiment recipe-retrieval. Each E10 child run contains its system label, frozen-manifest hash, metrics and saved test ranking. The parent run contains the aggregated evidence. Full-precision CSVs remain the numerical authority when printed tables round to four decimal places.

System	Train run	Validation run	E10 test run
E01	7565b0a06eca40d7ac2609c6191af7db	d9b7974ed0594d2582e87aa57906c87f	0e62fa799e2c424499517c63b67fac8b
E02	42f3905c5eea43368e08699bfb7146be	8edc932ae39248f6a658ac18f47cdb84	dbc9fa3d1b3e4a9dba4be67078448290
E03	b4e33b78d55948328d7ed0644d5bf8be	38645e8216dd42afa3bd8bdff2d78873	25fdd9bc3d1e4143978ae26d791bcc8d
LSA	29b6ca94a9d6453f90e32264e20ce425	4e4cf9e0ff5d46c087cff8266ec82923	845287a45f9e435b9bfabc0e3d2c814a
E04	0efff440125f441aab2f4bfb252f8b14	781dfb0e1a8b463cbd9628a21800e41f	8c26ccdf391d42f4b6af58084f6eef82
E05	4246c7020bbc4a4e8fc66b20f2a7b4df	c90c72b44d4f4c30b1002eb52dac7406	4adf8de413e84537961d122042208732
FT	90c9dfb87aff42668483433d8216ce2f	0444aa538a9948f8aecc25d5c42bb365	24091f7720a64f5089f3ceed1acfcb28

FT training-time provenance: 0437841237734100a7c1a9b26725d908; measured time 871.336 seconds imported from the historical CPU log, not a new training run. TF-IDF grouped CV run: dac82cb580e8403082cd9d0fabb40448. LSA CV run: d46efe09f7dd4835841f66b860d4d433. The CV JSONs give means, standard deviations, fold counts and corpus hashes. The master CSV links timing and CV fields to these sources.

The rubric checklist maps all five phases and their subrequirements to files, runs and responsible owners. It marks genuine regularization curves and real MLflow UI captures as missing for Hiếu, selected-BM25 clean-clone/container checks as pending for Hưng, and final human judgments as pending for the group. It also identifies incomplete student IDs, artifact-download configuration, clean-kernel notebook evidence and final presentation/rehearsal deliverables. It is an evidence map, not a guessed course grade.

The retrospective review of Hiếu's PR #20 records three findings: generated UI facsimiles and constant ablation loss arrays cannot establish measured experiments; scripted recommendation function calls are not naturally collected API user traffic; and experiment/model descriptions must match main FT E08, TF-IDF settings and 256-component LSA. Those findings are reported, not silently fixed or treated as satisfactory evidence. This branch contains Hoàng's work only.

References and primary artifacts: RecipeNLG dataset and source links (README Dataset); scikit-learn TF-IDF/TruncatedSVD/GroupKFold; rank-bm25; sentence-transformers MiniLM and MultipleNegativesRankingLoss; ranx metrics and paired comparison; MLflow tracking. Repository implementations are src/baseline.py, tune.py, finetune.py, evaluate.py and final_evaluation.py. Human-study conventions are locked by tests/test_ranking_metrics.py and annotation_guide.md. The companion Master Comparison CSV, frozen manifest and execution ledger are included in reports/final.